CHLORAMPHENICOL (EYE)

Bacterial conjunctivitis
Fridge storage (drops)

CHEMICAL NAME 2,2-dichloro-N-[(1R,2R)-1,3-dihydroxy-1-(4-nitrophenyl)propan-2-yl]acetamide
CLASS Broad-spectrum ophthalmic antibiotic (amphenicol)
TRADE NAMES Chloromycetin · Optrex Infected Eyes · Brochlor · Golden Eye Antibiotic

THERAPEUTIC USES / INDICATIONS

- Bacterial conjunctivitis.
- Superficial eye infections: blepharitis (ointment), keratitis (prophylaxis), corneal abrasion (prophylaxis).
- Post-operative prophylaxis.

PHARMACODYNAMICS

- Binds **50S ribosomal subunit** → inhibits bacterial protein synthesis (bacteriostatic); broad spectrum incl. *Staph.*, *Strep.*, *H. influenzae*.

ADMINISTRATION

- **Eye drops 0.5%:** wash hands, tilt head back, pull lower lid down, instil drop without touching eye.
- **Eye ointment 1%:** 1 cm ribbon into lower fornix at night (or 3–4 times daily alone).
- Remove contact lenses during treatment.
- Store unopened drops in fridge; discard 28 days after opening.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Transient stinging, burning.
- Blurred vision (ointment).
- Bitter taste (drainage into throat).

ADVERSE EFFECTS (SERIOUS)

- Hypersensitivity, contact dermatitis.
- Aplastic anaemia (extremely rare, idiosyncratic; reported with topical use).

DOSAGES

- **Drops:** 1 drop every 2 h for first 48 h, then every 4 h (while awake).
- **Ointment:** at night (with drops) or 3–4 times daily.
- Continue for 48 h after symptoms resolve (usually 5 days total).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity; personal/family history of blood dyscrasias (aplastic anaemia).
- Caution in pregnancy/breastfeeding (topical considered low risk).

KEY INTERACTIONS

- No significant interactions (topical).

PHARMACOKINETICS

- Good corneal penetration; minimal systemic absorption (trace).

NURSING MANAGEMENT

- Teach correct instillation; avoid contaminating dropper.
- Red flags needing urgent ophthalmology: eye pain, photophobia, reduced vision, contact lens wearer, neonates (ophthalmia neonatorum).
- Infection control (hand hygiene).

PATIENT EDUCATION

- Wash hands before and after; do not share towels.
- Use frequently for the first 2 days, then 4 times daily.
- No contact lenses until 24 h after finishing.
- See a doctor urgently for eye pain, light sensitivity or reduced vision.

KEY POINTS

- Broad-spectrum eye antibiotic.
- Continue 48 h after resolution.
- Drops every 2 h for 48 h, then 4-hourly.
- Red flags: pain, photophobia, vision loss.